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RESEARCH ARTICLE

Effective removal of the rhodamine B dye by citric acid-modified clay as adsorbent: kinetics, thermodynamics and adsorption isotherms

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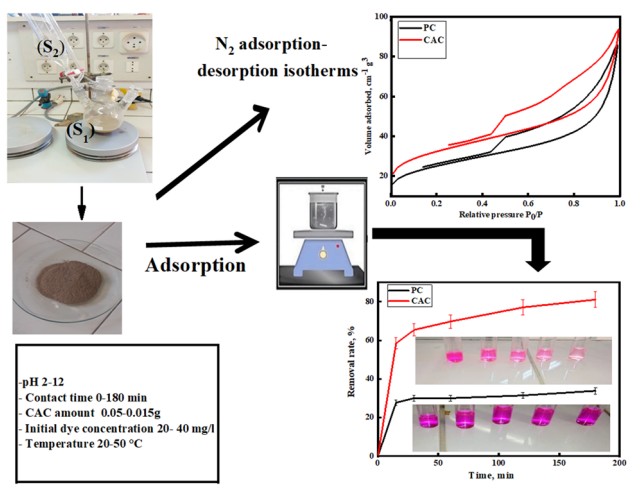
Abstract

Research on dye removal from wastewater using adsorbents is increasing. However, the development of suitable adsorbents and the assessment of their adsorbing capacities are still insufficient. Here, Citric Acid-modified Clay (CAC), a promising adsorbent, was prepared for removing the Rhodamine B (RhB) dye. Many techniques (X-ray diffraction, energy-dispersive X-ray spectroscopy, Brunauer–Emmett–Teller method, scanning electron microscopy, and thermogravimetric analyses) were employed to characterize the as-prepared adsorbent. The Brunauer–Emmett–Teller analysis showed that the surface area increased from 89 $m^2\ g^{-1}$ in purified clay to 114.50 $m^2\ g^{-1}$ in CAC. The effects of several operating parameters, including pH, temperature, adsorbent amount, contact time, and initial RhB concentration, were evaluated. Linear fitting favored Langmuir, while non-linear analysis indicated Freundlich, evidencing heterogeneous adsorption. The kinetic analysis demonstrated that the pseudo-second-order model best described RhB adsorption onto CAC. The thermodynamic parameters showed that RhB adsorption onto CAC was endothermic and spontaneous. These findings indicated that CAC is an efficient adsorbent for removing cationic dyes and could play a potential role in wastewater treatment by adsorption.

Graphical Abstract



Keywords Adsorption · Citric acid · Rhodamine B · Clay · Characterization

Introduction

In the last decade, the rise in textile production, particularly, the printing and dyeing processes, has caused an increase in dye consumption and consequently in wastewater dye

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concentration (Behera et al. 2024). Moreover, biological, analytical, and optical sciences make extensive use of Rhodamine B (RhB), a highly water-soluble basic dye of the xanthene class (Hou et al. 2011). RhB is also employed for paper printing, textile and leather dyeing, and by the food industry. However, its presence in environmental and food matrices can have severe carcinogenic, genotoxic, and neurotoxic effects, and it is an irritant of the respiratory system, eyes, and skin (de Morais Pinos et al. 2022). Therefore, as these effluents discharged by industries threaten human and aquatic life (Babaei et al. 2017; Li et al. 2024), they must be removed from contaminated wastewater using efficient, low-cost, and eco-friendly modified mineral materials as adsorbents using simple technologies. The existing modifiers negatively impacted water transmittance and plant photosynthesis. Searching for another eco-friendly modifier at mild conditions based on simple technology, without affecting dye removal effectiveness, seems to be an urgent task. This is a significant challenge considering the dye solubility, synthetic origins, complex aromatic structure, and biodegradability (Hou et al. 2011; Li et al. 2024).

Advanced treatment technologies, such as adsorption on activated carbon (Chauke et al. 2024; Gohr et al. 2022), thermolysis and coagulation (Lau et al. 2018), membrane separation (Cazzorla et al. 2018; Ye et al. 2018), electrochemical discoloration (Kusmierek et al. 2011), photocatalytic degradation (Fellah et al. 2022) and biological processes (BaniHani et al. 2019), have been employed for the elimination of organic pollutants. Nevertheless, due to their high operating and regeneration costs, these treatment techniques are not recommended. Adsorption on natural clay materials is a cost-effective, easy-to-apply, and effective technique compared to other methods (Chauke et al. 2024). Natural clays are widely utilized in water treatment due to their high specific surface area, promising adsorption capabilities, accessibility, swelling properties, and cation exchange capacity (He et al. 2022). A good adsorption capacity is mostly related to the adsorbent and adsorbate characteristics, but also influenced by other variables, such as pH, contact time, temperature, and initial adsorbate concentration. Several studies on mineral clays (montmorillonite) (Mahmoudian et al. 2018), zeolite (Alver and Metin 2012), kaolin, and bentonite (Mouni et al. 2018) have demonstrated their performance in dye removal from wastewater (Sarkar et al. 2024; Zhang et al. 2024). Modified clays are interesting adsorbent materials due to their porosity, high specific surface area, and ion exchange capacity, low cost, non-toxicity, and stability (Zhang et al. 2019). Particularly, Gan et al. (2012) found that Sepiolite modified with cetyltrimethylammonium bromide (CTAB/sepiolite) removed RhB more effectively than unmodified mineral clays. However, the high toxicity and cost of modifiers, as well as the complicated modification technologies, limit their use in the wastewater treatment field. Using toxic modifiers may result in more water contamination, where the human risks are emphasized. In fact, the use of organic surfactants (e.g., CTAB, HDTMA) can introduce toxicity or secondary pollution. Some modifiers may not be biodegradable or environmentally safe. Organic/inorganic acid treatment can increase the adsorption capacity of clay toward various types of dyes. However, the clay structure could readily be destroyed by strong inorganic acids such as sulphuric acid and hydrochloric acid. Moreover, the disposal of used modified clays, after their use as adsorbents, can pose waste management challenges, which in turn affects their dispersibility, especially in polymer matrices or suspensions. Modification techniques must be simple, so be, afterward, easily applied at the industrial scale at low prices. Surface functionalization may involve: (i) expensive reagents (e.g., quaternary ammonium compounds), (ii) multistep procedures (acid activation, intercalation, grafting), and (iii) toxic solvents or energy-intensive conditions. Therefore, it is crucial to find a mild and eco-friendly modifier to easily change natural clay without affecting the dye removal effectiveness.

Citric acid (CA), a non-toxic organic acid, is abundant and available at low cost when synthesized from natural biomass. In addition, the -OH and -COOH groups of CA are effective linking groups for its fixation on natural clays and RhB loading, and also efficient chelating groups for the removal of heavy metals and organic dyes. Some authors have already investigated CA-modified adsorbents for the adsorption of dyes and heavy metals (Wang et al. 2017; Yan et al. 2018). Research on hexavalent chromium (Cr (VI)) adsorption with magnetic citric acid bentonite (MCAB), performed by Wang et al. (2017), investigates the use of a modified bentonite clay for removing Cr (VI) from wastewater via a moderate, fast, and environmentally friendly method that avoids secondary pollution. This latter study focuses on a composite material, which combines the adsorption properties of bentonite with the magnetic properties of iron oxide and the functional groups of citric acid. According to recent studies performed by Zhang et al. (2019), raw bentonite (RB) was chemically modified by CA to obtain a low-cost and environment-friendly citric acid incorporated bentonite (CAB) adsorbent, for Congo Red (CR) adsorption as an anionic dye. It is worth noting that several clay minerals, such as bentonite (Beheshti et al. 2018; Wang et al. 2017; Zhang et al. 2019), montmorillonite (Mahmoudian et al. 2018), zeolite (Abdelrahman 2018; Boumnijel et al. 2023a, 2023b), and kaolin (Li et al. 2018), have been reported for the removal of dyes from wastewater. Nevertheless, smectite has not been considered for investigation using CA-modified clay as an adsorbent. Because smectite is an abundant clay in the south of Tunisia (El Hamma-Gabes, Tunisia) as a natural resource in this region, and CA is also available at low cost, CA-modified Clay (CAC) will be tested, in the present study, for RhB removal from aqueous

solutions, as a cationic dye. The as-prepared adsorbent was characterized using X-ray diffraction (XRD), energy-dispersive X-ray spectroscopy (EDX), Brunauer-Emmett-Teller (BET) method, scanning electron microscopy (SEM), and thermogravimetric analysis (TGA). Dye removal using CAC as adsorbent is a promising and innovative alternative solution. The effects of some operating parameters (temperature, pH, initial dye concentration, contact time, and adsorbent amount) on CAC adsorption capacities were assessed. Then, using the experimental findings, the kinetic, thermodynamic, and adsorption isotherm characteristics were also investigated.

Materials and methods

Materials

The raw clay (RC) used as an adsorbent in this paper was a grey clay from Djebel Aidoudi El Hamma-Gabes, Tunisia. The chemical composition was 50.8% SiO_2, 18.8% Al_2O_3, 8.5% Fe_2O_3, 2.3% CaO, 1.5% MgO, 1.4 92% K_2O, 2.2% Na_2O, and 1.9% P_2O_5 (Yahya et al. 2021). RhB (purity ≥ 99%; $C_{28}H_{31}ClN_2O_3$; Sigma Aldrich) molecular weight and maximum absorbance were 479.01 g mol^{-1} and 554 nm, respectively. CA ($C_6H_8O_7$) (purity ≥ 99%), sodium chloride (NaCl) (purity ≥ 99.5%), sodium hydroxide (NaOH) (purity ≥ 99%), and hydrochloric acid (HCl) (purity = 37%) were from Merck and were employed without additional purification.

Methods

CAC preparation

Clay fraction separation RC fractions were separated based on Stokes' Law according to the method described by Macías-Quiroga (Macías-Quiroga et al. 2018). RC (50 g) was dispersed in distilled water (5 L). After 20 min of homogenization using a magnetic stirrer, the obtained suspension was transferred to a graduated cylinder for gravitational deposition and left to stand for 16 h to obtain the clay fraction <2 µm. This clay fraction was selected due to its high mineralogical purity and greater surface reactivity. Then, the upper 20-cm fraction was removed. This process was repeated until the upper fraction became clear. Finally, the suspension was centrifuged at 4500 rpm for 15 min to precipitate the clay fraction that was dried at 60 °C, crushed, and sieved through a 100 mesh size. This fraction was denoted as purified clay (PC).

PC modification For PC modification, 5 g PC was dissolved in 50 ml of deionized water in a three-neck round-bottom flask, resulting in solution 1 (S1). Solution 2 (S2), made by dissolving 2.78 g CA in 50 ml of deionized water at room temperature, was added drop by drop to S1. The mixture aging was performed by agitation at 60 °C for 3 h. The resulting CA/PC was collected from the suspension by centrifugation (4500 rpm/15 min) and washed several times with distilled water to remove any remaining CA before drying in an oven at 60 °C for 24 h. The obtained CAC was ground, sieved through a 200 mesh size, and kept in a dryer until further analysis or use.

Characterization methods

To verify functional groups' modifications and structural characteristics of CAC, XRD patterns of PC and CAC were produced using a Bruker D5000 diffractometer (Netherlands) with nickel-filtered Cu Kα radiation ($\lambda = 0.15406$ nm) from 5 to 80° at a scan speed of 0.02°/s. The specific elemental species' concentrations were assessed by EDX, using an X-Max^N SDD Detector (Oxford Instruments). A field emission scanning electron microscope (Hitachi S4800, Instruments) was used to capture the surface morphology. An N_2 adsorption–desorption analyzer, with two Accelerated Surface Area and Porosimetry Systems (ASAP-2010 and ASAP-2020 Micromeritics), was used to measure the specific surface area (m^2 g^{-1}) with the BET method. A simultaneous Differential Scanning Calorimetry (DSC)-TGA instrument (SDT Q600, TA Instruments), from 25 °C to 1000 °C at a heating rate of 10 °C/min, was used for TGA analysis. A T80 UV/VIS Spectrophotometer was used to measure RhB concentrations.

RhB adsorption

Adsorption experiments were performed in batch mode by adding different adsorbent concentrations (from 0.01 to 0.03 g/l) to 250 ml Erlenmeyer flasks that contained 200 ml of RhB solution (20–40 mg/l), at different temperatures (20 °C to 50 °C), and different pH (2–9). Flasks were stirred using a magnetic stirrer at 600 rpm. NaOH or HCl solutions were used to adjust the pH of the RhB solutions. RhB adsorption kinetics were determined by stirring the RhB solution for 3 h until reaching equilibrium. At different time points, RhB samples were collected and filtered through a Micropore filter (0.2 µm). Then, samples were examined using a UV/VIS spectrophotometer (DR 2000, Hach Company) set to 554 nm to determine RhB concentration. At a given time (t), the amount of RhB adsorbed onto CAC (Q_e) (mg/g) was obtained as follows:

$$Q_e = \frac{(C_0 - C_t) * V}{m} \quad (1)$$

$$\text{Removal rate } (\%) = \frac{C_0 - C_e}{C_0} \quad (2)$$

where Q_e (mg/g) is the RhB adsorbed at equilibrium; C_0, C_t, and C_e (mg/l) are the aqueous RhB concentrations at time 0, at time t, and at equilibrium, respectively; V (l) is the aqueous solution volume; m (g) is the adsorbent weight.

It is worth noting that all studies were done in duplicate or triplicate, and the reported data are the means with a relative error of 3%.

Results and discussion

The as-prepared adsorbent was characterized. Additionally, the effect of various operating factors (adsorbent amount, contact time, RhB initial concentration, pH, and temperature) on CAC adsorption capacities were investigated. Then, using the experimental findings, the kinetic, thermodynamic and adsorption isotherm parameters were evaluated.

Characterization of PC and CAC

XRD analysis

Figure 1a shows the XRD patterns of PC and CAC. The d-spacing (d_{001}) of PC is 15.68 nm, corresponding to a typical montmorillonite pattern (Mukhopadhyay et al. 2019). After treatment with CA, the (d_{001}) value of CAC, assessed as 15.72 nm, is basically the same as PC, considering the combined instrumental and experimental error for a Bruker D5000 diffractometer. Accordingly, there is no obvious

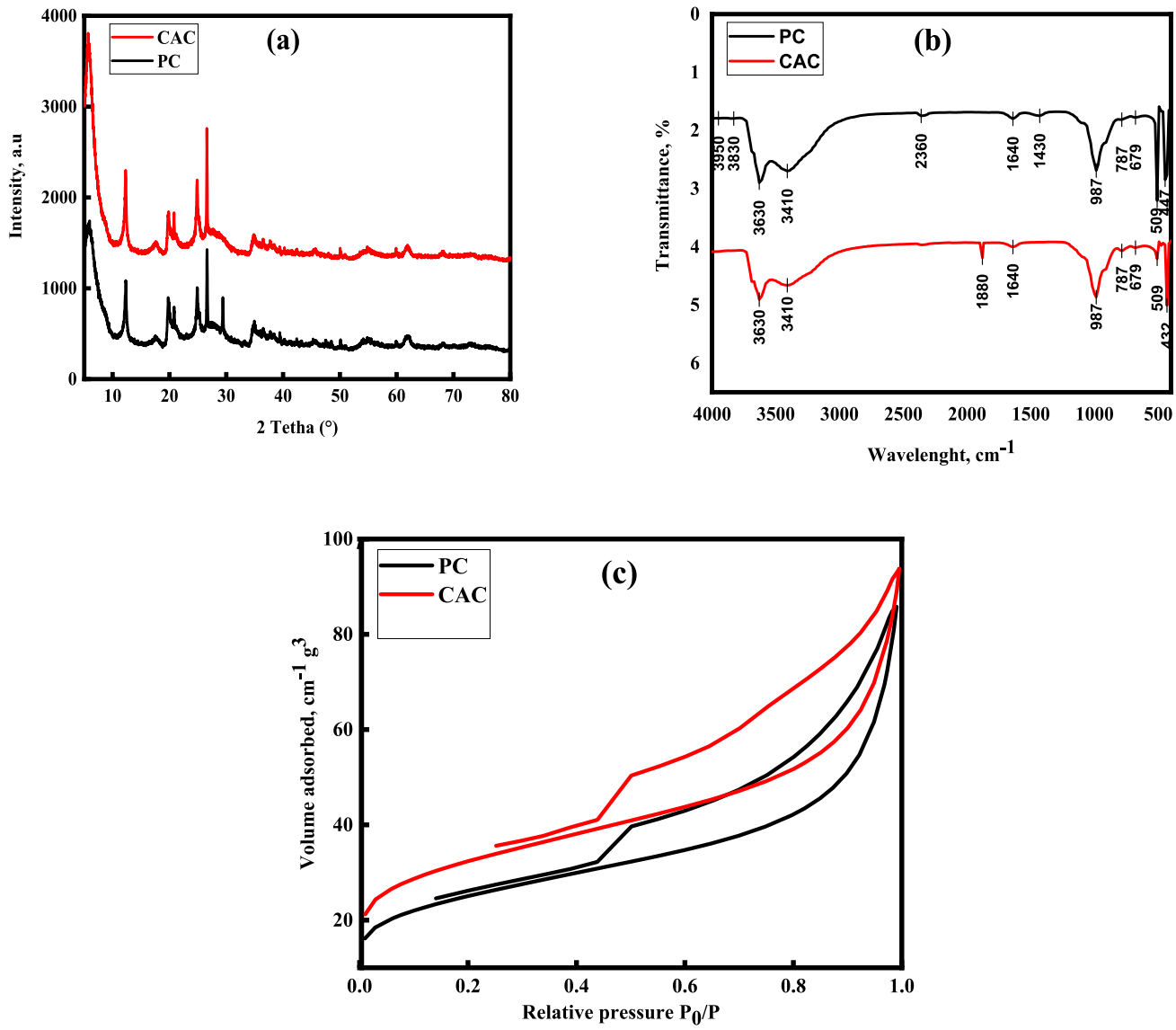


Fig. 1 **a** XRD patterns of the PC and CAC. **b** FT-IR spectra of PC and CAC. **c** N_2 adsorption–desorption isotherms of PC and CAC

change in CAC layer spacing after the introduction of CA, where the RhB molecules cannot penetrate the CAC interlayers during adsorption, due to their large dimensions (1.22 nm wide, 1.1 nm deep, and 0.5 nm thick) (Rao et al. 2020). Accordingly, RhB molecules are mostly adsorbed on the CAC surface. Moreover, CA molecules are exclusively adsorbed on the smectite surface and cannot be intercalated into its interlayer.

FT-IR analysis

Figure 1b shows the FT-IR spectra of PC and CAC. In the PC spectrum, the bands at 509 cm^{-1} and 987 cm^{-1} correspond to Si-O-Al and Si-O-Si stretching, respectively. Calcite ($CaCO_3$) vibration is associated with the band at 1430 cm^{-1}, whereas adsorbed water is responsible for the absorption band at 1640 cm^{-1} in PC. The bands at 3630 and 3410 cm^{-1} correspond to the -OH stretching vibration of adsorbed water and hydrogen bonding between water molecules and the Si–O surface, respectively (Gammoudi et al. 2012; Yahya et al. 2021). In the CAC sample, the new band at 1880 cm^{-1} was probably attributed to C = O symmetrical stretching, related to ester linkage formation, which may confirm the esterification reaction between PC (smectite) and CA. Given that PC is an inorganic alumino-silicate material with reactive hydroxyl groups and exchangeable cations, there are many -OH groups on its surface that may be esterified by the COOH functions of heated CA, forming an ester linkage (Segovia-Sandoval et al. 2018; Zhang et al. 2019). As mentioned by Perera (2022), the stretching frequency value of the C = O bond in the ester molecule is between 1820 and 1770 cm^{-1}, slightly lower than 1880 cm^{-1}, but certainly confirming the esterification reaction. The observed difference depends on the inductive and resonance effects exerted by the atoms/groups attached to the C = O, related to the carboxyl groups of CA (Perera 2022). Significant reductions in the broad -OH stretching vibration intensities around 3200–3830 cm^{-1} are observed, confirming the esterification reaction of PC and CA. Furthermore, CA activity at high temperatures may explain the disappearance of the calcite band at 1430 cm^{-1} after its dissolution by releasing calcium and carbonate ions (Ren et al. 2025).

BET and adsorption–desorption isotherms

The N_2 adsorption–desorption isotherms of PC and CAC (Fig. 1c) had a type H_3 hysteresis loop related to slit-shaped pores (Macías-Quiroga et al. 2018). After treatment with CA, the specific surface area increased from 89.00 m^2 g^{-1} in PC to 114.50 m^2 g^{-1} in CAC, and the total pore volume increased from 0.11 cm^3 g^{-1} (PC) to 0.12 cm^3 g^{-1} (CAC) (Table 1). Moreover, the mean pore diameters were 4.80 nm (PC) and 4.30 nm (CAC). The increase in the CAC surface area is promoted by the CA treatment, possibly through the dissolution of impurities, leaching of structural cations, and the removal of several minerals, which creates additional micro/mesopores (Essekri et al. 2022). SEM images and EDX analysis can later verify this suggestion. In this context, Kong et al. (2014) reported that the specific surface areas of the tested clay minerals increased after reacting with organic acids, specifically oxalic and citric acids.

TGA/DSC analysis

The TGA/DSC curves of PC and CAC (Fig. 2) showed that thermal stability was higher in PC than in CAC, when taking into account their total mass loss (15.72% for CAC and 12.57% for PC). The TGA curve of PC shows five decomposition phases (Fig. 2a). The first two mass losses (3.79% and 0.66%), occurring between room temperature (~20 °C) and 230 °C, can be explained by the loss of free water molecules in the interlayers or at the clay surface. The third mass loss (6.64%), between 250 and 620 °C, corresponds to smectite dehydroxylation. The fourth mass loss (0.95%) at 664.43 °C is related to calcium carbonate ($CaCO_3$) decomposition. The last mass loss (weight loss of 0.52%), between 750 and 1000° C, is attributed to decarboxylation (Nawel et al. 2020). The DSC curve for CAC (Fig. 2b) is almost identical to that of PC. The main differences are the amounts of mass loss between 0 and 300 °C and the disappearance of the peak at 664.43 °C, which corresponds to the decomposition of $CaCO_3$ as a result of heating and CA action (Fernandes et al. 2020). In effect, the citric acid/clay decomposition process in CAC is observed at 200–300 °C, accompanied by an increased weight loss (7.78% against 3.79%), about 3.99% compared to PC, proving the successful adsorption of CA on the PC structure (Silva et al. 2008; Takahashi et al. 2000). These findings are in agreement with decomposition studies of CA/silica composites, where the citric acid decomposition process was observed at around 300 °C (Takahashi et al. 2000).

SEM and EDX analyses

SEM was used to determine the surface morphology of PC and CAC (Fig. 3). CAC has a more open, porous, and fluffy

Table 1 Textural characteristics of PC and CAC

Adsorbent	BET surface area (m^2 g^{-1})	Total pore volume (cm^3 g^{-1})	Pore diameter (nm)	BJH* (cm^3 g^{-1})
PC	89.00	0.11	4.80	0.11
CAC	114.50	0.12	4.30	0.12

*Pore size distribution calculated with the Barrett-Joyner-Halenda (BJH) method

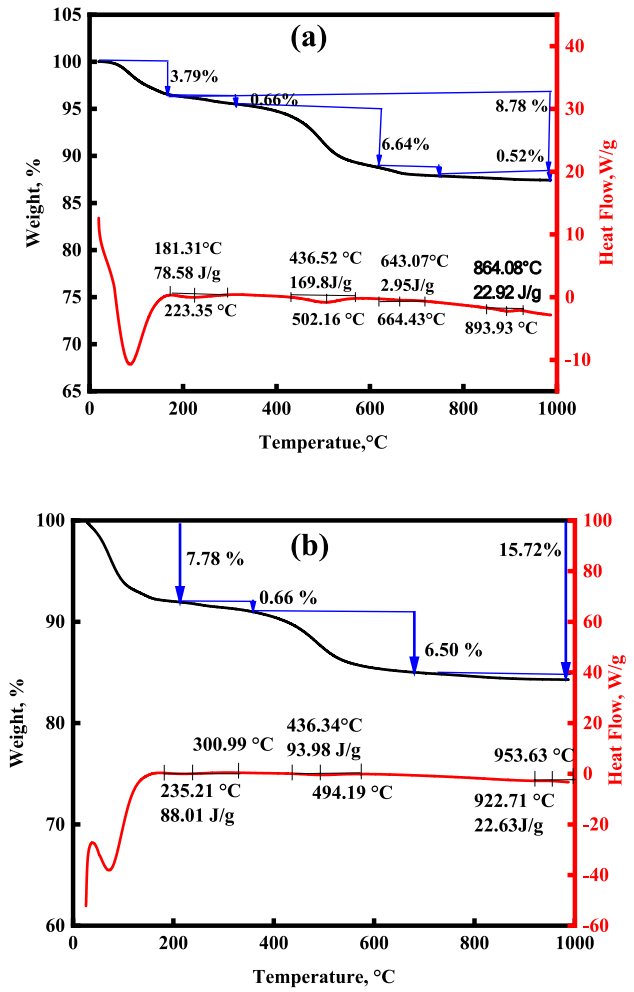


Fig. 2 TGA/DSC patterns of **a** PC and **b** CAC

structure than PC (Fig. 3a, b). CAC becomes rougher, with a few small flaky scales on its surface. The observed changes in CAC microstructure (*vs*. PC) suggest that CA can dissolve impurities and bind to the PC surface. This, as previously demonstrated by the BET data, contributes to increasing the specific surface area and total pore volume.

Then, EDX was used to assess the elemental composition of PC and CAC (Fig. 3c,d). Carbon represented 10.70 wt% in CAC and only 5.80 wt% in PC, confirming the successful grafting of organic CA molecules (Zhou et al. 2015). It is well known that Ca ions are present in the interlamellar layer, whereas Mg and Fe ions are found in the octahedral sheets (Amari et al. 2018; Kooli and Yan 2013). Considering the effect of their cationic exchanges, Ca, Mg, and Fe ions have a negative impact on PC adsorbent capacity. In this context, CA can increase PC adsorbent capacity by decreasing Ca, Mg, and Fe ions contents by 53.33, 20.00, and 21.87%, respectively, as seen in the EDX spectrum of CAC (Fig. 3c,d). Upon treatment with CA, the slight decrease in Si content could be attributed to the partial breakdown of the octahedral layers and to the reduction of Fe, Mg, and Ca content in CAC (Kong et al. 2014). The decrease in Ca content can be ascribed to the simple dissolution of Ca ions by protons due to their location in the interlamellar region (Novikau and Lujaniene 2022). From these results, it can be concluded that CAC, effectively incorporated on the PC surface, enhances the textural characteristics of CAC for wastewater purification purposes.

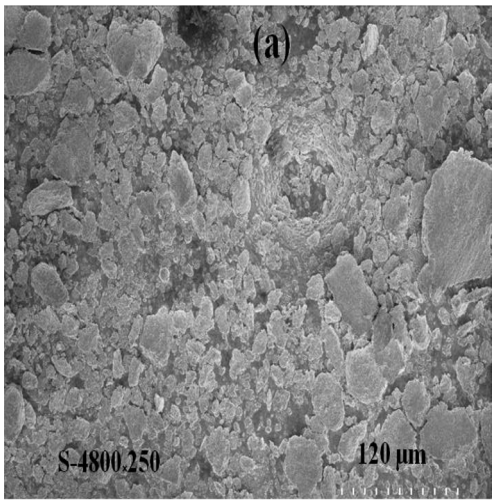
Adsorption tests

Preliminary study of RhB adsorption using PC and CAC

First, the effect of the contact time on RhB adsorption (initial concentration of 20 mg/l) onto PC and CAC (0.12 g/l) was evaluated at ambient temperature (20 °C) and pH = 2 (Fig. 4). RhB removal rate increased with the contact time until reaching equilibrium. With PC as adsorbent, the RhB removal rate was 27.74% after 15 min and then gradually increased to 33.8% after 30 min. The highest RhB reduction rate (81.11%) was observed using CAC at 180 min. This improvement is possibly due to the creation of new active sites and the increase in the specific surface area upon treatment with CA (Amari et al. 2018; Novikau and Lujaniene 2022), as previously shown by the BET and SEM analyses. Zhang et al. (2019) reported similar adsorption kinetics for Congo red with a removal efficiency of 87%. Moreover, by comparing RhB removal efficiency using PC and CAC, we can conclude that CAC has faster reaction kinetics and therefore can be considered very cost-effective for industrial applications.

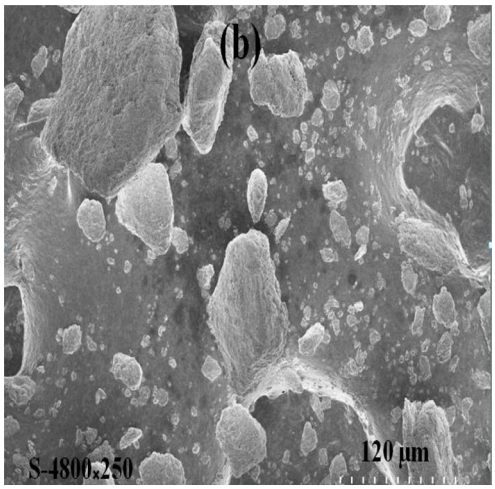
RhB adsorption by CAC

To investigate RhB removal efficiency using CAC, adsorption was monitored at different pH values (from 2 to 9) (Fig. 5a). Rhodamine B adsorption by CAC is closely related to pH. So, RhB adsorption decreased with increasing pH values. At low pH values, it is assumed that both chemisorption and physisorption could occur, leading to greater RhB removal. Nevertheless, at higher pH values, the decrease in RhB removal can be attributed to the reduction of electrostatic attraction between cationic RhB and the protonated adsorption sites of the CAC. In fact, at low pH, the CAC surface is partially protonated ($-OH/-OH_2^+$ /neutralized carboxyls (–COOH)), and RhB tends to stay in its cationic form (RhB^+). Despite that, adsorption still increases at low pH because the protonated surface makes some carboxyl groups less negatively charged, which minimizes repulsion from excess negative sites, allowing other forces, namely hydrogen bonding, π–π stacking and van der Waals with the remaining negatively charged functional groups on the CAC ($-COO^-$), to act more effectively.



(c)

Element	Weight (%)
C	5.80
O	48.70
Na	0.30
Mg	1
Al	10.80
Si	22.50
Cl	0.10
K	1.40
Ca	1.50
Ti	0.50
Fe	9.60
Total	100



(d)

Element	Weight (%)
C	10.70
O	45.60
Na	0.10
Mg	0.80
Al	10.60
Si	22.00
K	1.27
Ca	0.70
Ti	0.70
Fe	7.50
Total	100

Fig. 3 SEM images of **a** PC and **b** CAC materials. EDX results for **c** PC and **d** CAC

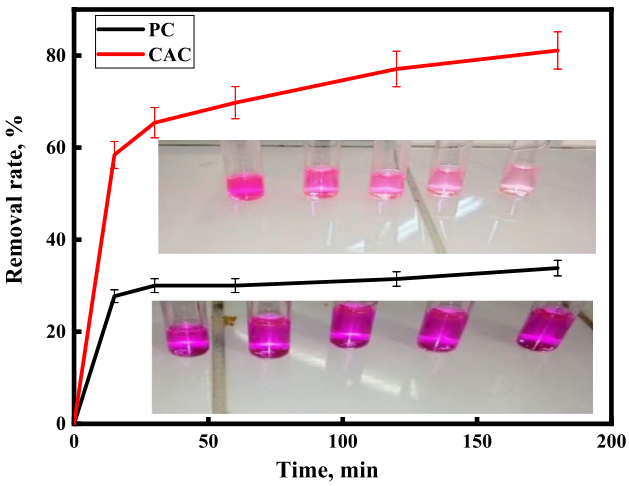


Fig. 4 Effect of contact time on RhB adsorption using PC and CAC (T = 20 °C, pH = 2, C_0 = 20 mg/l, clay amount = 0.12 g/l)

At higher pH, the CAC surface, which is more negatively charged (–COO^-/–O^-), can attract RhB^+. However, RhB itself gradually becomes neutral (lactone and zwitterionic forms) at high pH (pH > 8), so electrostatic attraction decreases along with π–π interactions; therefore, the RhB adsorption process is reduced. Overall, the adsorption of RhB at acidic pH occurs predominantly through physisorption, driven by strong electrostatic attraction and hydrogen bonding. In addition, chemisorption may also contribute, mainly through chelation between the –COOH groups of the CAC and the amino group of RhB, along with π–π interactions between their aromatic structures. Nevertheless, Gad and El-Sayed (Gad and El-Sayed 2009) have reported, during their study on the removal of Rhodamine-B from aqueous solution by developing an efficient adsorbent from agricultural by-product with H_3PO_4 (or KOH) activation, that chemisorption dominates at low pH whereas chemisorption along with physisorption occurs at higher pH ranges.

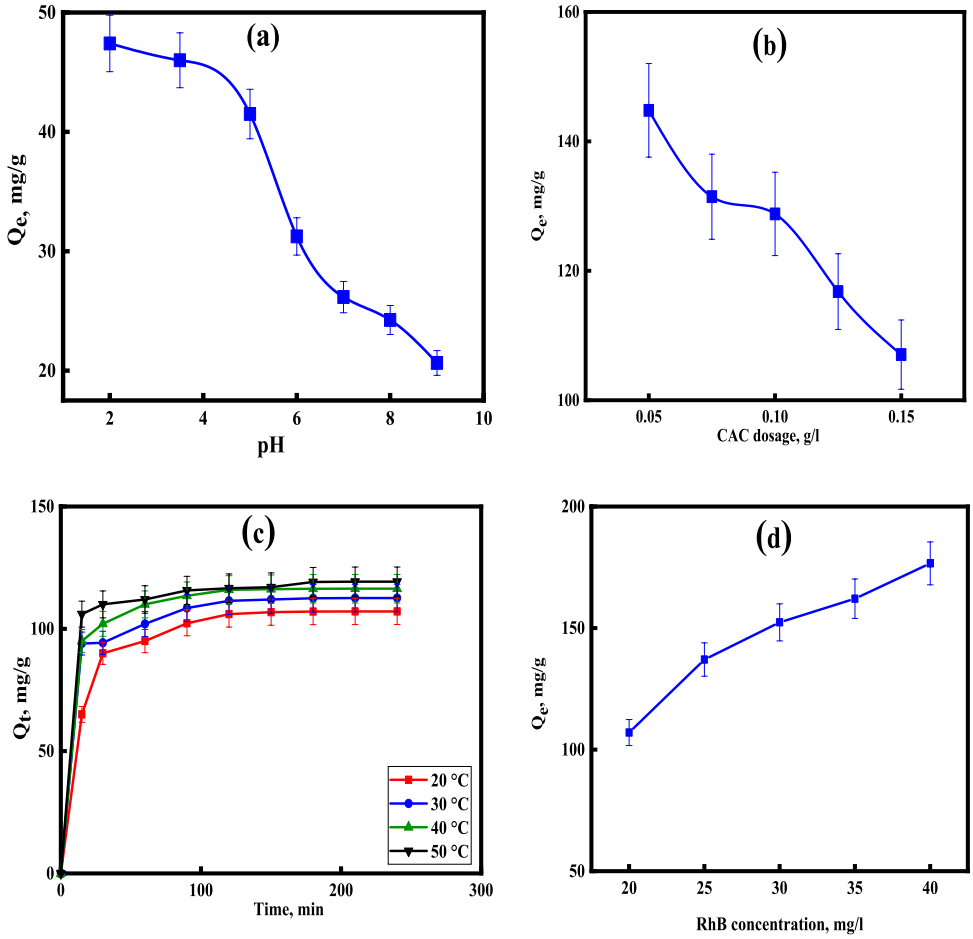


Fig. 5 **a** Effect of pH on RhB adsorption onto CAC (T = 20 °C, C_0 = 20 mg/l, clay amount = 0.12 g/l). **b** Effect of CAC amount on RhB adsorption (T = 20 °C, pH = 2, C_0 = 20 mg/l). **c** Effect of temperature on RhB adsorption onto CAC (pH = 2, C_0 = 20 mg/l, CAC amount = 0.12 g/l). **d** Effect of the initial RhB concentration on its adsorption onto CAC (T = 20 °C, pH = 2, CAC amount = 0.12 g/l)

Additionally, RhB adsorption is controlled by the diffusion processes. RhB can exist in three protonated forms: zwitterions ($RhB^{\pm}$) (pH > 4.0), monocationic species (RhB^{+}) (pH 1.0–3.0), and dicationic species (RhB_2^{2+}) (pH < 1.0) (Hou et al. 2011). At pH < 4, RhB remains mainly in its cationic and monomeric form, which can readily diffuse into the porous structure of CAC, resulting in higher adsorption. In contrast, at higher pH values, RhB adopts a zwitterionic form and tends to be in dimer structure, promoting aggregation into larger molecular clusters that hinder penetration into the CAC pores (Alhogbi et al. 2021; Gad and El-Sayed 2009). Moreover, electrostatic interactions between carboxylate groups of CAC and the xanthene moiety of zwitterionic RhB further enhance dimerization, leading to polymerization and precipitation. This process obstructs pore channels and ultimately decreases the adsorption capacity (Alhogbi et al. 2021; Lin et al. 1999; Lu et al. 2020). Considering the aforementioned results, the next experiments were performed at pH 2.

Using different CAC amounts (from 0.05 to 0.15 g) (Fig. 5b) showed that RhB adsorption decreased with increasing CAC amounts. This can be attributed to the unsaturation of adsorption sites that contribute to the decrease of adsorption uptake.

To identify the best temperature for RhB adsorption, experiments were carried out at different temperatures (from 20 to 50 °C) (Fig. 5c). RhB adsorption increased from 107.1 mg/g at 20 °C to 119.3 mg/g at 50 °C, indicating that RhB adsorption is a spontaneous endothermic process. Therefore, RhB adsorption on the CAC surface is facilitated at high temperatures due to the enhanced mobility of the large dye ions that may get sufficient energy to interact with active sites. As the adsorption capacity mostly depends on the chemical interactions between the adsorbate and functional groups on the adsorbent surface, it should increase with the temperature. Furthermore, the clay internal structure may swell at higher temperatures, enabling larger RhB molecules to penetrate more deeply (Imanipoor et al. 2021).

The effect of RhB initial concentration (from 20 to 40 mg/l) was also examined (Fig. 5d). The removal rate progressively increased with higher RhB concentrations. Higher initial dye concentrations may provide the required driving force to overcome the resistance to RhB molecule mass

transfer between the liquid and solid phases. Furthermore, increasing the initial dye concentration enhances the interaction between RhB and CAC. Importantly, as RhB concentration increased, the time needed to reach the equilibrium also increased. This can be explained by the theory that molecules must penetrate the boundary layer before being successfully diffused and adsorbed on the CAC surface. In addition, RhB agglomeration may occur at high concentrations, slowing down the adsorption process. Accordingly, it will require more time to reach the equilibrium with higher RhB concentrations. In conclusion, 20 mg/l seems to be the optimal RhB concentration for effective adsorption (Wierzbicka et al. 2022).

Compared with other adsorbents, the as-prepared CAC exhibits the highest adsorption capacity for RhB (Fig. 6) (Das et al. 2006; Zhang et al. 2021). For further comparative study, the experimental conditions of different sources are specified in Table 2. Thus, CAC could be adopted by many industries as an eco-friendly and economical adsorbent for treating the industrial dye-contaminated wastewater.

Adsorption isotherm studies

The Langmuir, Freundlich, and Temkin models (Baccar et al. 2013) were used to describe the CAC adsorption mechanism based on the experimental data (Fig. 7a–-c). In each model, different parameters were used to predict the adsorption mechanisms.

The Langmuir isotherm, shown in its linear form by Eq. 3, is used to describe the monolayer adsorption on a uniform surface with a limited number of identical sites (Baccar et al. 2013).

$$\frac{C_e}{Q_e} = \frac{1}{K_L * Q_m} + \frac{1}{Q_m} * C_e \tag{3}$$

where Q_m (mg/g) is the theoretically estimated maximum adsorption capacity, K_L (l/mg) is the Langmuir isotherm constant relating to the adsorbate/adsorbent interaction energy, C_e (mg/l) is the solution concentration at equilibrium, and Q_e (mg/g) is the adsorbent uptake at equilibrium.

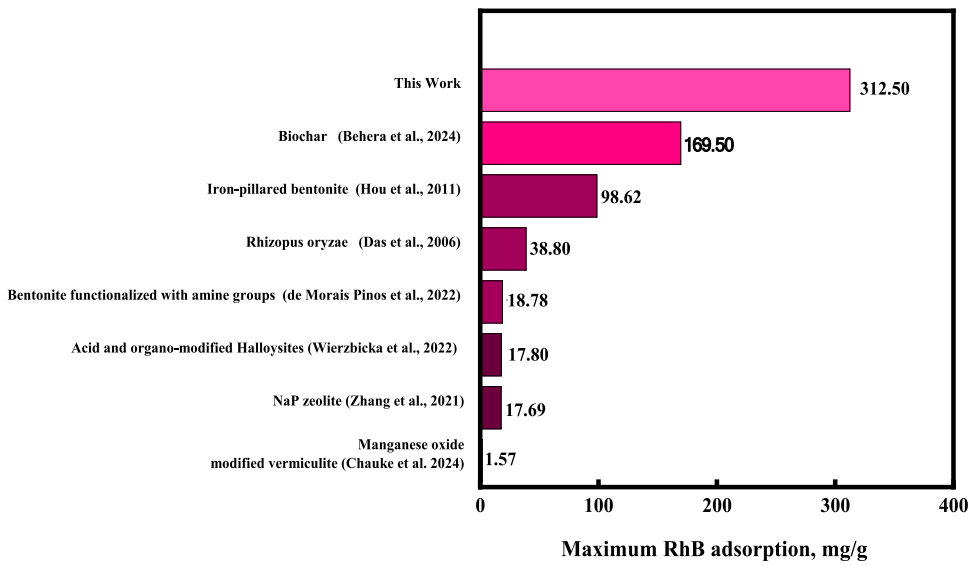


Fig. 6 Maximum RhB adsorption by previously described adsorbents and CAC

Table 2 Experimental conditions of different sources for the maximum RhB adsorption values

	PH	Contact time (min)	Temperature, K	Clay dosage, g/l
This work	2	180	323.15	0.12
Biochar (Behera et al. 2024)	2	25	308	1.20
Iron-pillared bentonite (Hou et al. 2011)	7	30	398	30.00
Rhizopus oryzae Biomass (Das et al. 2006)	7	360	303	0.10
Bentonite functionalized with amine groups (de Morais Pinos et al. 2022)	-	120	298	0.05
Acid- and Organo-Modified Halloysites (Wierzbicka et al. 2022)	2	180	298	0.667
NaP zeolite (Zhang et al. 2021)	6	60	298	2.50
Manganese oxide modified vermiculite (Chauke et al. 2024)	10	45	303	1.75

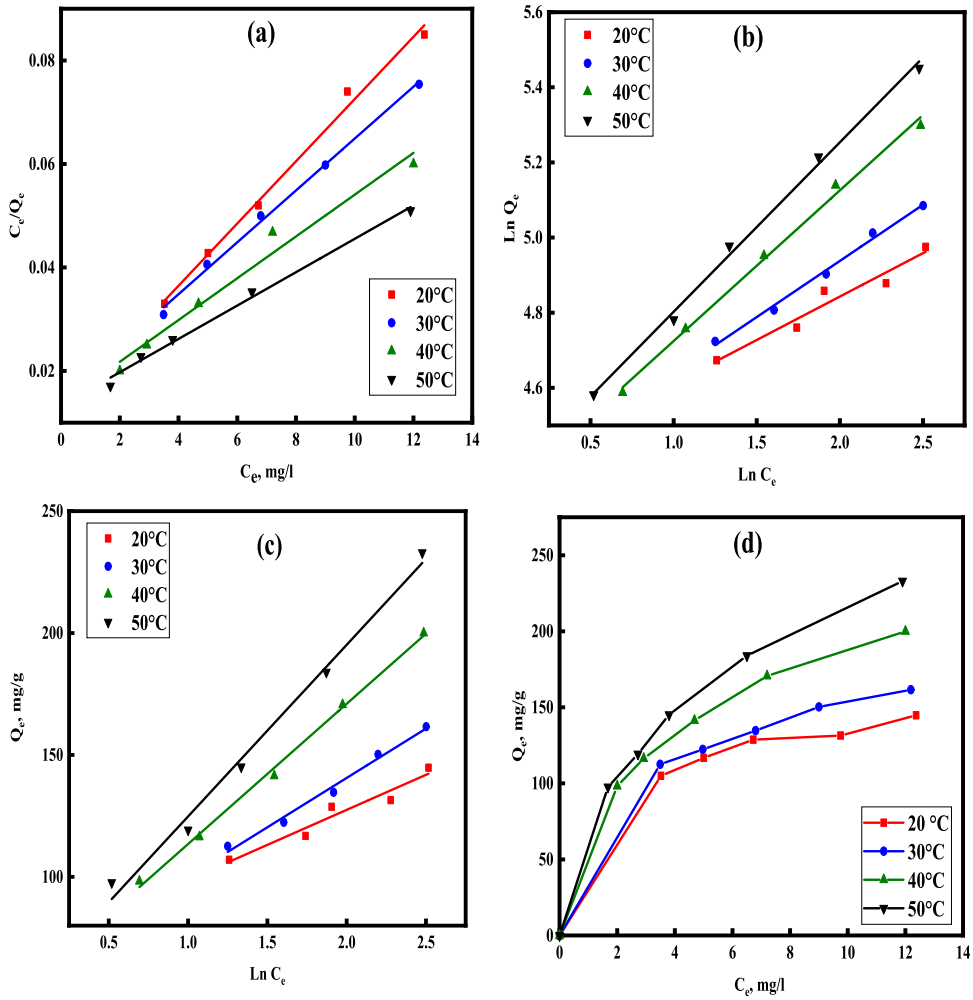


Fig. 7 **a** Langmuir, **b** Freundlich, and **c** Temkin models for RhB adsorption on CAC at different temperatures. **d** Q_e as a function of C_e at different temperatures (pH = 2, C_0 = 20 mg/l, clay amount = 0.12 g/l)

The dimensionless constant R_L (Eq. 4), defined as the equilibrium parameter, specifies whether an isotherm is irreversible ($R_L = 0$), favorable ($0 < R_L < 1$), linear ($R_L = 1$), or unfavorable ($R_L > 1$).

$$R_L = \frac{1}{1 + K_L C_0} \tag{4}$$

where C_0 (mg/l) is the initial concentration of the solute in the solution.

The linear expression of the Freundlich model (Eq. 5) supposes a homogenous adsorption (Lin et al. 2018; Zhang et al. 2019):

$$LnQ_e = LnK_F + \frac{1}{n} LnC_e \tag{5}$$

where n and K_F (mg/g) are the Freundlich constants related to the adsorption intensity and the adsorption capacity, respectively. The heterogeneous factor (1/n) can be used to express the affinity between adsorbent and adsorbate (Baccar et al. 2013).

According to Eq. 6, the Temkin isotherm is usually used in its linear version.

$$Q_e = B_t LnK_t + B_t LnC_e \tag{6}$$

$$B_t = \frac{RT}{b} \tag{7}$$

where B_t (J/mol) is the Temkin constant related to the adsorption heat, K_t (L/g) is the Temkin isotherm constant, R (J/mol K) is the gas constant (8.314 J/mol K),b is the Temkin isotherm constant (J/mol). and T is the absolute temperature (K) (Wierzbicka et al. 2022).

To understand the mechanism of RhB adsorption onto CAC, the experimental data were fitted to the aforementioned equilibrium isotherm equations. The different isotherm parameters, obtained from the plots (Fig. 7), are presented in Table 3. The linear correlation coefficient, R^2, of each isotherm equation, when fitted to the experimental data, is used. On comparison among these models, the Langmuir isotherm is a good fit for experimental

Table 3 Isotherm model parameters of RhB adsorption on CAC at various temperatures

	Langmuir isotherm				Freundlich isotherm			Temkin isotherm		
T (°C)	Q_m(mg/g)	K_L(L/g)	R_L	R²	K_F(mg/g)	1/n	R²	K_t (L/g)	B_t (kJ/mol)	R²
20	166.60	0.480	0.096	0.993	73.690	0.230	0.946	11.45	28.730	0.940
30	200	0.338	0.127	0.996	76.700	0.298	0.993	4.360	40.470	0.987
40	250	0.285	0.138	0.977	75.100	0.400	0.996	2.640	57.530	0.997
50	312.50	0.240	0.164	0.993	77.470	0.450	0.996	2.160	70.400	0.991

RhB adsorption data (Fig. 7a and Table 3), because, independently of the applied temperature, the R_L values were between 0 and 1. Furthermore, its R^2 correlation values ($0.977 < R^2 < 0.996$), as seen in Table 3, are much higher than those of the Freundlich ($0.945 < R^2 < 0.996$) and Temkin ($0.940 < R^2 < 0.987$) isotherms. The adsorption isotherms had an L-type shape (Fig. 7d) in accordance with the classification by Giles et al. (1974). The L-shaped Langmuir isotherm, which shows no significant competition between solvent and adsorbent or adsorbate sites, is indicative of monolayer adsorption. However, RhB adsorption on CAC is an endothermic process because the monolayer adsorption capacity (Q_m) increases from 166.60 to 312.50 mg/g with the solution temperature increase (Table 3). Typically, the Freundlich constant K_F is positively correlated with adsorption capacity. In this study, K_F slightly increases with rising temperature, suggesting that higher temperatures can enhance the adsorption capacity of CAC. The preferential adsorption on CAC is also confirmed by the values of the heterogeneity factor (1/n), which remains between 0 and 1 at all tested temperatures. Adsorption is generally considered difficult when 1/n is greater than 1.0, while values below 0.5 indicate favorable adsorption (Lu et al. 2020). In this case, all 1/n values are less than 0.5, demonstrating that RhB could be fairly adsorbed onto CAC. Furthermore, the increase in 1/n with temperature suggests that elevated temperatures facilitate the adsorption process. The positive values of the constant Bt, related to the adsorption heat in the Temkin isotherm, suggest an endothermic reaction (Baccar et al. 2013). RhB adsorption on CAC is better explained by the Langmuir isotherm based on the linear method results.

Nonlinear models should be applied when analyzing isotherm and kinetic parameters for adsorption investigations, since they offer a more accurate representation of the experimental data compared to linear techniques. By avoiding the distortions caused by linearization and directly fitting the original model equation to the data, nonlinear regression analysis yields more accurate parameter estimations. Accordingly, a non-linear approach for isotherm models assessment is conducted. The estimated isotherm parameters with their non-linear correlation coefficients, obtained using the Origin software, are depicted in Table 4. From this table, the non-linear correlation coefficient value (r^2) is larger for the Freundlich isotherm, as compared to the Langmuir and Temkin isotherms. The adsorption data fitted well with the Langmuir isotherm in its linearized form, suggesting the possible occurrence of monolayer adsorption. However, when analyzed using nonlinear regression, the Freundlich model provided a better fit, indicating surface heterogeneity and the coexistence of multiple adsorption sites. Since nonlinear fitting avoids the bias introduced by linearization, the Freundlich model is considered more reliable in describing the RhB adsorption onto CAC.

Adsorption kinetics

To predict the predominant adsorption mechanism and the adsorption rate, RhB adsorption kinetics on CAC were modeled. Diffusion phenomena and two kinetic models, namely the Lagergren's pseudo-first order and pseudo-second order models, were examined (Juan et al. 2015; Zhang et al. 2019). The Lagergren's pseudo-first order kinetic model, for adsorption in solid/liquid systems, is represented by Eq. 8 (Wang and Guo 2020).

$$\mathrm{Ln}\left(Q_e - Q_t\right) = \mathrm{Ln}Q_e - K_1 t \tag{8}$$

Table 4 Non-Linear fit adsorption isotherm parameters for RhB adsorption

	Langmuir isotherm			Freundlich isotherm			Temkin isotherm		
T (°C)	Q_m (mg/g)	K_L (L/g)	R²	K_F (mg/g)	1/n	R²	K_t (L/g)	B_t (kJ/mol)	R²
20	162.73	0.53	0.996	81.58	0.223	0.955	29.35	10.57	0.997
30	196.30	0.35	0.996	76.48	0.300	0.991	40.47	4.36	0.998
40	253.45	0.29	0.997	76.82	0.390	0.994	57.53	2.65	0.999
50	309.06	0.24	0.996	78.88	0.441	0.996	70.40	2.17	0.996

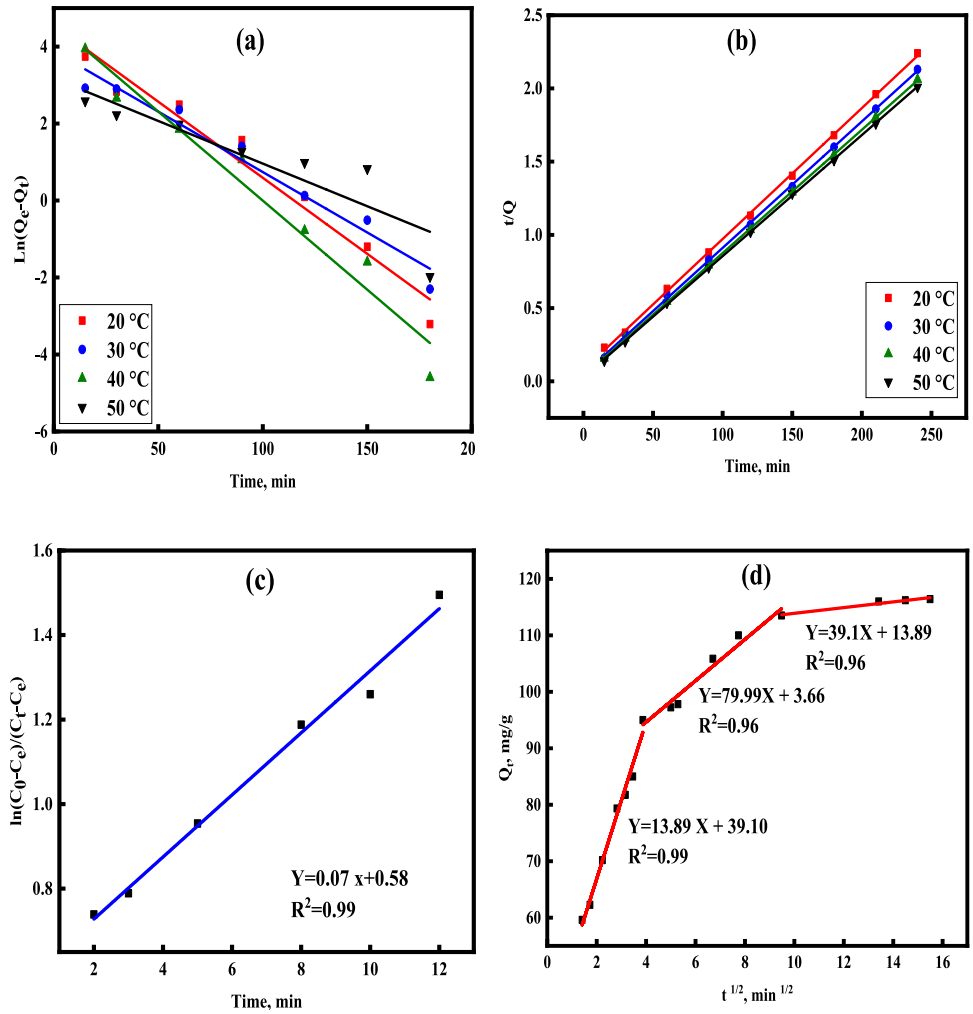


Fig. 8 **a** Pseudo-first order kinetic model of RhB adsorption onto CAC at different temperatures (pH = 2, C_0 = 20 mg/l, CAC amount = 0.12 g/l). **b** Pseudo-second order kinetic model of RhB adsorption onto CAC at different temperatures (pH = 2, C_0 = 20 mg/l, CAC amount = 0.12 g/l). **c** External diffusion model of RhB adsorption onto CAC (pH = 2, C_0 = 20 mg/l, CAC amount = 0.12 g/l, T = 40 °C). **d** Intra-particle diffusion model of RhB adsorption onto CAC (pH = 2, C_0 = 20 mg/l, CAC amount = 0.12 g/l, T = 40 °C)

where Q_e and Q_t (mg/g) are the RhB adsorbed amounts at equilibrium and at time t (h), respectively, and K_1 (min^{-1}) is the adsorption rate constant.

To further understand RhB adsorption kinetics on CAC, a pseudo-second order model (Zhang et al. 2019) was used to fit the experimental data (Eq. 9):

$$\frac{t}{Q_e} = \frac{1}{K_2 Q_e^2} + \frac{1}{Q_e} t \tag{9}$$

where K_2 is the adsorption rate constant of the pseudo-second order model (g mg^{-1} min^{-1}). At 20 °C, plotting (t/Q_e) vs t yields the K_2 values.

The diffusion phenomenon describes the molecule flow from the solution to the solid surface. Generally, two diffusion models are mostly used: the intra-particle diffusion model, which can be the limiting step in the adsorption process, and the external diffusion model. The experimental findings were used to adjust the mathematical models to validate the extent of each diffusion mode's contribution to the overall process. The linear version of the external diffusion model is provided by Eq. 10 (Belaid and Kacha 2011):

$$\mathrm{Ln}\frac{(C_0 - C_e)}{(C_t - C_e)} = K\frac{a}{v} = K\prime t \tag{10}$$

where C_e, C_0, and C_t (mg/l) are the solute concentration at equilibrium, at time 0 and time t, respectively; *a* is the solid/liquid interface area; *v* (l) is the solution volume; and K' (min^{-1}) is the inter-particle diffusion rate constant.

To determine the internal diffusion mechanism, an intra-particle diffusion model (Eq. 11) based on the theory proposed by Weber and Morris was used:

$$Q_t = K_p * t^{\frac{1}{2}} + I \tag{11}$$

where I is the intercept and K_p (mol $min^{-1/2}$ g^{-1}) the intra-particle diffusion rate constant obtained from the slope of Q_t versus $t^{1/2}$ (Wierzbicka et al. 2022; Zhang et al. 2019).

Figure 8 shows the linear forms of the pseudo-first-order (Fig. 8a) and pseudo-second-order models (Fig. 8b) of RhB

adsorption onto CAC at various temperatures. The kinetic model parameters, computed from the best fitting plots, are listed in Table 5, which suggest that whatever the adopted temperature, the linear correlation coefficients of the pseudo-second-order model are very high ($R^2 = 0.999$) and the computed equilibrium adsorption capacity (Q_{ecal}) values are in good agreement with the experimental data (Q_{eexp}). Therefore, the pseudo-second-order kinetic model seems to be more accurate than the pseudo-first-order model, suggesting that chemical adsorption plays a crucial role in the overall process (Bazan-Wozniak et al. 2024). The adsorption rate constant K_2 values of the pseudo-second-order model increased with the adsorbed amounts at equilibrium (Q_e), likely due to the faster RhB adsorption and the shorter time to reach the equilibrium state. Consequently, higher temperatures promote the adsorption process.

The diffusion processes are most likely to be rate-limiting steps that control adsorption (Belaid and Kacha 2011). To further investigate the adsorption kinetics, the external and intra-particle diffusion models were studied during RhB adsorption onto CAC at 40 °C. The plot of $\ln\frac{(C_0-C_e)}{(C_t-C_e)}$ vs t should therefore be assessed to verify whether the external diffusion step is determined for the adsorption process (Fig. 8c). As seen in Fig. 8c, mass-transfer resistance surrounding the particles was not important in the early stages of adsorption considering that the plot line did not pass through the origin, confirming that external diffusion is not the only limiting step in RhB adsorption onto CAC. In general, the intra-particle diffusion model occurred into three stages: (1) the diffusion of the adsorbate to the external surface of the adsorbent through interparticle spaces, i.e., the boundary layer diffusion. (ii) The gradual adsorption stage where the intra-particle diffusion rate is regarded as the rate-limiting parameter and lastly (3) the adsorption equilibrium stage. As seen in Fig. 8d, the fitting plot of Q_t versus $t^{1/2}$ could be divided into three stages where RhB molecules diffuse rapidly on the CAC external surface, then penetrate into its interior surface and lastly reach adsorption equilibrium. In short, the RhB adsorption process on CAC could be divided into two steps namely, the boundary layer diffusion and the intra-particle diffusion. The corresponding plots do not pass through the origin, demonstrating that intra-particle diffusion is involved in the adsorption process; despite it is not the only rate-controlling step. Thus, both internal and external diffusion controlled RhB adsorption onto CAC. The same findings were reported by Lu et al. (2020) during their study on RhB adsorption from aqueous solution using Goat Manure Biochar. Likewise, this is similar to a report on the adsorption of RhB dye from Wastewater onto activated carbon from Palm Tree Fiber Waste (Alhogbi et al. 2021).

Thermodynamic studies

The temperature effect on RhB adsorption was also investigated because thermodynamic parameters, such as enthalpy changes (ΔH°), entropy changes (ΔS°), and Gibbs free energy changes (ΔG°), provide insights into the spontaneity and heat changes occurring during adsorption (Baccar et al. 2013). To this aim, experiments were performed using an initial RhB concentration of 20 mg/l, 30 mg of adsorbent, and a temperature range from 20 to 50 °C.

Equations 12, 13, and 14 were used to calculate the thermodynamic parameters:

$$K_d = \frac{Q_e}{C_e} \tag{12}$$

$$\Delta G^\circ = -RTLnK_d \tag{13}$$

$$LnK_d = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{RT} \tag{14}$$

where R is the ideal gas constant (8.314 J mol^{-1} K^{-1}), T is the absolute temperature (K), and K_d is the distribution coefficient for adsorption.

The thermodynamic parameters (ΔH°, ΔS°, ΔG°), computed from Fig. 9a, are collected in Table 6. As the temperature increased, the ΔG° values decreased, suggesting more effective adsorption at higher temperatures (Table 6). Moreover, the ΔG° negative values demonstrate the spontaneous nature of RhB adsorption and the feasibility of the adsorption process for the adsorbent (Lamrani et al. 2023). The decrease of ΔG° with temperature shows that high temperatures can

Table 5 Kinetic model parameters and correlation coefficients during RhB adsorption onto CAC

		Pseudo-first order kinetic model			Pseudo-second order kinetic model		
T (°C)	Q_{eexp} (mg/g)	Q_{ecal} (mg/g)	K_1 (min^{-1})	R^2	Q_{ecal} (mg/g)	K_2 (g mg $^{-1}$ min^{-1})	R^2
20	107.10	93.69	0.0395	0.964	111.85	0.00101	0.999
30	112.60	47.94	0.0314	0.962	116.27	0.00150	0.999
40	116.40	101.49	0.0462	0.959	119.14	0.00204	0.999
50	119.30	23.80	0.0221	0.805	120.48	0.00257	0.999

Q_{ecal}: calculated equilibrium adsorption capacity

Q_{eexp}: experimental adsorption capacity

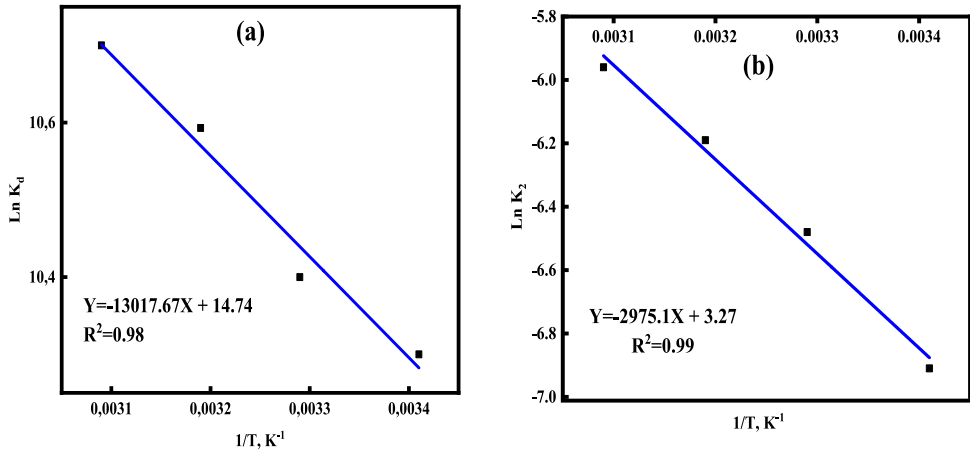


Fig. 9 **a** Plot of Ln K_d versus 1/T. **b** Plot of Ln K_2 versus 1/T during RhB adsorption onto CAC (pH = 2, C_0 = 20 mg/l, CAC amount = 30 mg, T = 20 °C)

Table 6 Thermodynamic parameters at various temperatures

T (°C)	ΔG° (kJ/mol)	ΔH° (kJ/mol)	ΔS° (J/mol.K)	Ea (kJ/mol)	R²
20	−25.10	10.87	122.50	24.19	0.98
30	−26.26				
40	−27.57				
50	−28.75				

promote the spontaneous trend of adsorption, which is consistent with the previous results obtained from the Freundlich model. Further evidence that the adsorption process involves both physical interactions and partial chemical bonding, commonly referred to as mixed adsorption, is provided by the ΔG° values, which fall within the range of −20 to −40 kJ·mol^{-1}. A positive ΔH° value indicates that the process would be endothermic (Lamrani et al. 2023; Lu et al. 2020). The positive ΔS° value indicates that the adsorption is an entropy-increasing process, and the solid–liquid interface will become more random during adsorption (Lin et al. 2018), reflecting the affinity of the adsorbent material towards RhB (Gad and El-Sayed 2009). Therefore, an increase in randomness at the interface between the CAC and the rhodamine B solution results in a higher efficiency of dye removal.

As the pseudo-second order kinetic model described well the RhB adsorption process onto CAC, the activation energy was computed using the Arrhenius equation (Eq. 15) and the rate constant K_2 (Baccar et al. 2013):

$$LnK_2 = LnA - \frac{E_a}{RT} \tag{15}$$

where K_2, A, E_a, R, and *T* are the rate constant of the pseudo-second order model, Arrhenius factor, activation energy, gas constant, and temperature, respectively.

The obtained activation energy value (Ea = 24.19 kJ/mol), computed from the slope of the Ln K_2 versus 1/T plot (Fig. 9b), indicates that RhB adsorption onto CAC is predominantly governed by physisorption, as the value is below 40 kJ/mol (Sheikhhosseini et al. 2014). However, its magnitude, combined with the ΔG° values (−20 to −40 kJ/mol), suggests the contribution of stronger interactions such as hydrogen bonding or chelation, supporting the occurrence of a mixed adsorption mechanism. Furthermore, the positive values of Ea reveal that the RhB adsorption rate increases with temperature. Likewise, as enthalpy change (ΔH_0) and activation energy (Ea) have the same sign, RhB adsorption onto CAC is endothermic (Sheikhhosseini et al. 2014).

Mechanism of RhB adsorption onto CAC

During the modification of purified clay (PC), citric acid (CA) introduces carboxylic acid (–COOH) groups onto the clay surface through esterification or surface complexation with surface hydroxyl (–OH) groups (e.g., Si–OH and Al–OH). As a result, the citric-acid-modified clay (CAC) surface contains a high density of –COOH groups in addition to existing –OH groups, significantly altering its adsorption behavior compared to raw or purified clay. Depending on solution pH, some –COOH groups deprotonate into –COO^-, with greater deprotonation occurring above pH 3–4. RhB adsorption onto CAC can proceed through three main mechanisms: (i) chelation/complex formation, (ii) stabilization via hydrogen bonding and π–π interactions, and (iii) pore-filling fixation. The chelation mechanism arises from interactions between surface carboxylate groups and RhB. RhB exists mainly as a cationic dye (RhB^+) due to its quaternary ammonium group within the xanthene ring system. Negatively charged –COO^- groups on CAC electrostatically attract RhB^+ and may coordinate with its amine nitrogen atoms. Two chelation pathways are possible: (a) direct chelation, where –COO^- donates lone pairs to form coordination bonds with the electron-deficient RhB nitrogen, and (b) metal-mediated chelation, where CA's –COOH/–COO^- groups first chelate metal cations (Al^{3+}, Fe^{3+}, Ca^{2+}) in the clay matrix,

creating bridging complexes that subsequently bind RhB. This latter mechanism is stronger and more chemisorptive than simple electrostatic adsorption. FTIR results (Fig. 10) further support chelation between –COO⁻ groups of CAC and RhB nitrogen atoms. Additional stabilization occurs through hydrogen bonding between residual –COOH/–OH groups and RhB functional groups ((–NR_2) and –C = O), as well as π–π stacking interactions between aromatic domains of RhB and CAC(Bazan-Wozniak et al. 2024). Finally, pore-filling contributes to adsorption: RhB molecules diffuse into CAC micropores and mesopores, where they can become physically trapped. This diffusion-driven fixation is consistent with adsorption kinetic analysis, particularly the intra-particle diffusion model. Altogether, RhB adsorption onto CAC is best described as a mixed mechanism, involving electrostatic attraction, coordination bonding, hydrogen bonding, π–π interactions, and pore-filling effects.

In the FTIR spectrum of CAC after RhB adsorption (CAC–RhB) (Fig. 10), the characteristic bands at 3630 and 3410 cm^{-1}, corresponding respectively to the –OH stretching vibrations of adsorbed water and hydrogen bonding between water molecules and surface Si–OH groups, became slightly shifted and broadened. This broadening indicates stronger hydrogen bonding interactions between RhB, surface hydroxyl groups, and interfacial water molecules. In addition, the intensity change observed at 3630 cm^{-1} may also reflect chelation or interaction of –OH groups from the octahedral sheets (Al–OH/Mg–OH) in the clay framework. A new band at 3390 cm^{-1} is assigned to the N–H stretching vibration, confirming the contribution of RhB. Likewise, the appearance of a new band at 2980 cm^{-1}, associated with aliphatic C–H stretching of the RhB ethyl groups, further confirms dye loading. New absorption peaks at 1420 and 1340 cm^{-1} are attributed to the symmetric and asymmetric

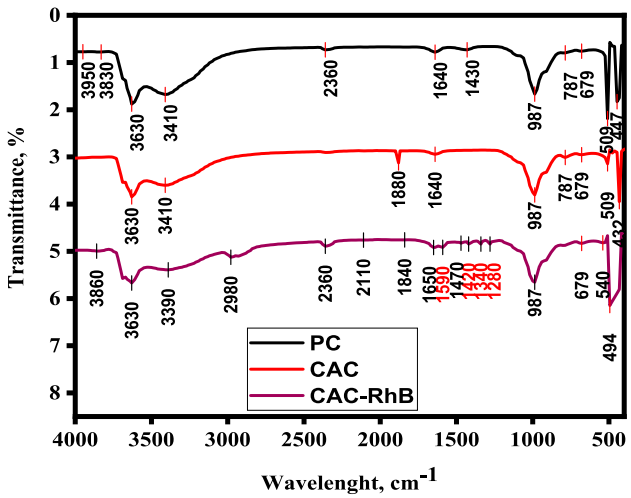


Fig. 10 FT-IR spectra of PC, CAC and RhB adsorbed onto CAC (CAC-RhB)

stretching vibrations of carboxylate (–COO^-) groups (Rossi et al. 2018; Zhang et al. 2019), which can electrostatically interact with the positively charged amine nitrogen of RhB, suggesting the involvement of carboxylates in coordination. A distinct band at 1590 cm^{-1}, assigned to C = C ring stretching, evidences the presence of the RhB aromatic xanthene system on CAC. The new band at 1280 cm^{-1}, corresponding to C–N stretching in a chelation-sensitive region, indicates electron delocalization at the nitrogen atom. The broad band at 987 cm^{-1}, related to Si–O stretching vibrations of Si–O–M (M = Al or Si), became slightly broadened after surface modification and dye adsorption. Finally, the appearance of a new broad band at 494 cm^{-1}, attributed to metal–oxygen (M–O) or metal–oxygen–carbon (M–O–C) vibrations, supports the involvement of metal-ion bridging. Overall, the thermodynamic (ΔG°, Ea) results indicate adsorption is predominantly physisorptive. At the same time, FTIR reveals specific interactions (e.g., carboxylate coordination/metal–ion bridging), indicating a mixed mechanism with non-negligible chemisorption contributions, where the functional groups –NR_2, aromatic –C = O, –OH, and –COOH act as the principal active adsorption sites.

Conclusion

CAC, an effective adsorbent, was synthesized by modifying purified smectite (PC) with citric acid (CA) for RhB removal. XRD confirmed that CA molecules were mainly adsorbed on the PC surface without interlayer intercalation, while FTIR revealed ester linkages responsible for chelation with RhB. N_2 adsorption–desorption isotherms showed an increase in surface area, likely due to new pore formation, whereas TGA/DSC indicated lower thermal stability of CAC compared to PC. SEM micrographs further supported surface binding of CA, enhancing textural properties. CAC achieved a maximum adsorption capacity of 312.5 $mg \cdot g^{-1}$ for RhB at pH 2 and 50 °C. Isotherm analysis showed Langmuir capacities increased with temperature (166.60 → 312.50 $mg \cdot g^{-1}$), with linear fitting favoring Langmuir and non-linear analysis favoring Freundlich, indicating heterogeneous adsorption. Kinetic studies demonstrated that the pseudo-second-order model best described the process, governed initially by external mass transfer and later by intra-particle diffusion. Thermodynamic parameters confirmed an endothermic and spontaneous adsorption. RhB adsorption onto CAC proceeds mainly via physisorption, with significant chemisorption contributions involving –NR_2, –C = O, –OH, and –COOH functional groups. Overall, CAC shows strong potential as a low-cost adsorbent for dye removal, and future work should explore its application to other persistent organic pollutants.

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Author contributions All authors contributed to the experimental study performed in the Research Laboratory Processes, Energetic, Environment and Electric Systems (PEESE) (ENIG). Material preparation, data collection, and analysis were performed by Imen Fellah, Ibtissem Boumnijel, Hedi Ben Amor, and Mahmoud Abid. The first draft of the manuscript was written by Imen Fellah; then, Ibtissem Boumnijel and Mikhael Bechelany commented on and corrected previous versions of the manuscript. All authors read and approved the final manuscript.

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Declarations

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to participate All authors are aware of and accept responsibility for the manuscript.

Consent for publication No part of the manuscript contents has been published or has been accepted for publication elsewhere.

Conflict of interest The authors declare that they have no conflict of interest.

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